



**Luis Lopez**

[luislopezdiaz@cmail.carleton.ca](mailto:luislopezdiaz@cmail.carleton.ca)

[carleton.ca/APEX](http://carleton.ca/APEX)

Department of Mechanical and Aerospace Engineering  
Ottawa, Canada

Fall 2026

# MAAE 2300: Fluid Mechanics I

Problem Set 1

# Quick reference

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# Governing equations

## Thermodynamics & elasticity

**Density:**  $\rho = \frac{m}{V}$

**Specific gravity:**  $SG = \frac{\rho}{\rho_w}$

**Ideal gas law:**  $P = \rho RT$

**Bulk modulus:**  $E_v = -\frac{\Delta P}{\Delta V/V} = \rho \frac{\Delta P}{\Delta \rho}$

**Pressure scale:**  $P_{\text{abs}} = P_{\text{atm}} + P_{\text{gauge}}$

## Viscosity & surface tension

**Newton's viscosity:**  $\tau = \mu \frac{du}{dy}$

**Kinematic viscosity:**  $\nu = \frac{\mu}{\rho}$

**Droplet pressure:**  $\Delta P = \frac{2\gamma}{R} = \frac{4\gamma}{D}$

**Bubble pressure:**  $\Delta P = \frac{4\gamma}{R} = \frac{8\gamma}{D}$

**Capillary force:**  $F_\sigma = \gamma L \cos \theta$

# Problem 1

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## Problem 1: statement

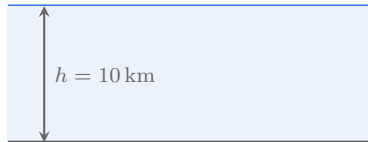
The pressure of the water at a depth of 10 km in the ocean is about 100 MPa (sea level atmospheric pressure is about 0.1 MPa). Assuming a value of the bulk modulus of elasticity  $K = 2100$  MPa ( $E_v$ ), what is the density of the water at that depth, if the S.G. (specific gravity) is 1.03 at the surface?

**(Ans:  $1081 \text{ kg m}^{-3}$ )**

# Problem 1: solution (part 1)

## ➔ Step 1: Sketch & given data

Sea surface:  $P_1 = 0.1 \text{ MPa}$ ,  $SG_1 = 1.03$



Ocean trench ( $h = 10 \text{ km}$ ):  $P_2 = 100 \text{ MPa}$

- Net pressure jump:

$$\Delta P = P_2 - P_1 = 99.9 \text{ MPa}$$

- Surface water density ( $SG_1 = 1.03$ ):

$$\rho_1 = SG_1 \rho_w = 1.03(1000) = 1030 \text{ kg m}^{-3}$$

- Bulk modulus of water:  $K = 2100 \text{ MPa}$  ( $E_v$ )

**Target:** water density at oceanic depth  $\rho_2$ .

## ➔ Step 2: Governing laws & assumptions

**Bulk modulus of elasticity ( $K$  or  $E_v$ ):**

$$K = -\frac{dP}{dV/V} = \rho \frac{dP}{d\rho} \approx -\frac{\Delta P}{\Delta V/V}$$

**Physical assumptions:**

- **Closed control mass:** A parcel of mass  $m$  is compressed as it descends; mass is strictly conserved:

$$m = \rho_1 V_1 = \rho_2 V_2 = \text{const}$$

- **Constant bulk modulus:**  $K \approx \text{const}$  over the pressure range  $\Delta P$ .

- **Temperature variation:** Isothermal/near-constant temperature is assumed in the deep layer.

# Problem 1: solution (part 2)

## ➔ Step 3: Algebra & symbolic derivation

From the definition of bulk modulus on parcel  $V_1$ :

$$K \approx -\frac{\Delta P}{\Delta V/V_1} \Rightarrow \frac{\Delta V}{V_1} = -\frac{\Delta P}{K}$$

The compressed volume at depth 2 is:

$$V_2 = V_1 + \Delta V = V_1 \left(1 - \frac{\Delta P}{K}\right)$$

Equating initial and final mass ( $m = \rho_1 V_1 = \rho_2 V_2$ ):

$$\rho_2 = \frac{\rho_1 V_1}{V_2} = \frac{\rho_1 V_1}{V_1(1 - \Delta P/K)}$$

Closed-form symbolic density formula:

$$\rho_2 = \frac{\rho_1}{1 - \frac{\Delta P}{K}}$$

## ✓ Step 4: Calculate & sanity check

**Numerical substitution:**

$$\begin{aligned} \rho_2 &= \frac{1030 \text{ kg m}^{-3}}{1 - \frac{99.9 \text{ MPa}}{2100 \text{ MPa}}} = \frac{1030}{1 - 0.04757} \\ &= \frac{1030}{0.95243} = \mathbf{1081.4 \text{ kg m}^{-3}} \approx \mathbf{1081 \text{ kg m}^{-3}} \end{aligned}$$

*Physical sanity check:*

- Hydrostatic compression yields  $\rho_2 > \rho_1$  (+4.87%).
- Even at 10 km depth (100 MPa  $\approx$  1000 atm), water compresses by only  $\sim 5\%$ , confirming that liquids are nearly incompressible in everyday engineering.
- Units ( $\text{kg m}^{-3}$ ) correct. ✓

# Problem 2

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## Problem 2: statement

The helium-filled blimp below is used at various athletic events. Determine the number of pounds of helium within it if its volume is  $68,000 \text{ ft}^3$  and the temperature and pressure are  $80^\circ\text{F}$  and  $14.2 \text{ psi}$ , respectively.

**(Ans: 668 lb)**



Figure: Blimp schematic.

## Problem 2: solution (part 1)

### ➔ Step 1: Sketch & given data

#### Given parameters (helium envelope):

- Enclosed volume:  $V = 68,000 \text{ ft}^3$
- Temperature:  $T = 80^\circ\text{F}$
- Pressure:  $P = 14.2 \text{ psi}$
- Working gas: Helium (He)
- Molar mass:  $M = 4.003 \text{ lbm/lbmol}$

**Target:** total mass of helium  $m$  in pounds (lbm).

### ➔ Step 2: Governing laws & assumptions

#### Ideal gas equation of state:

$$P = \rho RT \implies m = \rho V = \frac{PV}{RT}$$

#### Assumptions & physical rules:

- Helium behaves as an ideal gas at atmospheric conditions.
- Uniform  $T$  and  $P$  throughout the envelope.
- **Critical rule:** Absolute temperature and pressure must be used in the ideal gas law:

$$T \text{ in } ^\circ\text{R}, \quad P \text{ in lbf/ft}^2$$

## Problem 2: solution (part 2)

### ➡ Step 3: Algebra & symbolic derivation

Conversion to absolute English Engineering units:

- Absolute temperature ( $^{\circ}\text{R}$ ):

$$T = 80 + 459.67 = 539.67^{\circ}\text{R} \approx 540^{\circ}\text{R}$$

- Absolute pressure ( $\text{lbf}/\text{ft}^2$ ):

$$P = 14.2 \frac{\text{lbf}}{\text{in}^2} \times 144 \frac{\text{in}^2}{\text{ft}^2} = 2044.8 \frac{\text{lbf}}{\text{ft}^2}$$

- Specific gas constant for Helium:

$$R_{\text{He}} = \frac{R_u}{M} = \frac{1545.3}{4.003} = 386.0 \frac{\text{ft} \cdot \text{lbf}}{\text{lbm} \cdot ^{\circ}\text{R}}$$

### ✓ Step 4: Calculate & sanity check

Mass substitution:

$$m = \frac{PV}{RT} = \frac{(2044.8 \text{ lbf}/\text{ft}^2)(68,000 \text{ ft}^3)}{(386.0 \text{ ft} \cdot \text{lbf}/(\text{lbm} \cdot ^{\circ}\text{R}))(540^{\circ}\text{R})}$$

$$= 667.6 \text{ lbm} \approx 668 \text{ lb}$$

Physical sanity check:

- Density  $\rho_{\text{He}} \approx 0.0098 \text{ lbm}/\text{ft}^3 \ll \rho_{\text{air}} \approx 0.075 \text{ lbm}/\text{ft}^3$ .
- Displaced air weight is  $\sim 5100 \text{ lb}$ , yielding a net buoyant lift of  $\approx 4400 \text{ lbf}$  for payload and gondola. ✓

# Problem 3

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## Problem 3: statement

An aircraft carries an oxygen cylinder to which a pressure gauge is attached. The gauge reads 100 kPa(g). The air inside the aircraft is pressurized to 8 psi above the ambient pressure outside the aircraft, which is flying at 12,000 m in a standard atmosphere.

What is the absolute pressure of the cabin air and of the oxygen in the tank?

**(Ans: 74.56 kPa, 174.56 kPa)**

## Problem 3: solution (part 1)

### ➡ Step 1: Sketch & given data

Given flight parameters:

- Flight altitude:  $z = 12,000$  m
- Ambient pressure (ISA table):

$$p_{\text{outside}} = 19.40 \text{ kPa}$$

- Cabin pressure differential:

$$\Delta p_{\text{cabin}} = 8 \text{ psi}$$

- Tank gauge reading:

$$p_{\text{gauge,tank}} = 100 \text{ kPa(g)}$$

**Target:** absolute cabin pressure  $p_{\text{cabin,abs}}$  and tank pressure  $p_{\text{tank,abs}}$ .

### ➡ Step 2: Governing laws & assumptions

Pressure datum definition:

$$P_{\text{abs}} = P_{\text{reference}} + P_{\text{gauge}}$$

**Crucial physical observation:** The oxygen cylinder gauge is located **inside the aircraft cabin**, so its reference datum is the cabin air pressure ( $P_{\text{reference}} = p_{\text{cabin,abs}}$ ), not the outside ambient air!

- Cabin air is referenced to outside ambient pressure.
- Oxygen cylinder is referenced to pressurized cabin air.

## Problem 3: solution (part 2)

### ➔ Step 3: Algebra & symbolic derivation

Unit conversion to SI:

$$\Delta p_{\text{cabin}} = 8 \text{ psi} \times 6.8948 \frac{\text{kPa}}{\text{psi}} = 55.16 \text{ kPa}$$

Absolute pressure relations:

$$p_{\text{cabin,abs}} = p_{\text{outside}} + \Delta p_{\text{cabin}}$$

$$p_{\text{tank,abs}} = p_{\text{cabin,abs}} + p_{\text{gauge,tank}}$$

### ✓ Step 4: Calculate & sanity check

Numerical evaluation:

$$p_{\text{cabin,abs}} = 19.40 + 55.16 = \mathbf{74.56 \text{ kPa}}$$

$$p_{\text{tank,abs}} = 74.56 + 100 = \mathbf{174.56 \text{ kPa}}$$

*Physical sanity check:*

- Cabin pressure of 74.56 kPa corresponds to an effective cabin pressure altitude of  $\sim 2400 \text{ m}$  (8000 ft).
- This is the standard regulatory commercial limit for passenger oxygenation without supplemental masks. ✓

# Problem 4

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### Stokes–Oseen drag formula

According to the Stokes–Oseen formula, the drag force  $F$  on a sphere of diameter  $D$  in a fluid stream of low velocity  $V$ , density  $\rho$ , and viscosity  $\mu$  is given by:

$$F = 3\pi\mu DV + \frac{9\pi}{16}\rho V^2 D^2$$

Is the formula dimensionally consistent? If yes, prove it. If no, what should be the dimensions of the numerical constants to make it consistent?

**(Ans: Yes, dimensionally consistent)**

## Problem 4.1: solution

### ➔ Step 1 & 2: Primary dimensions

Express all physical variables in primary  $[M, L, T]$  dimensions:

- Drag force:  $[F] = MLT^{-2}$
- Dynamic viscosity:  $[\mu] = ML^{-1}T^{-1}$
- Flow velocity:  $[V] = LT^{-1}$
- Sphere diameter:  $[D] = L$
- Fluid density:  $[\rho] = ML^{-3}$
- Pure numbers:  $[3\pi] = [9\pi/16] = 1$  (dimensionless)

### ➔ Step 3: Dimensional evaluation of terms

**Term 1 (viscous Stokes drag):**

$$[3\pi\mu DV] = (ML^{-1}T^{-1})(L)(LT^{-1}) = MLT^{-2}$$

**Term 2 (inertial Oseen correction):**

$$\left[\frac{9\pi}{16}\rho V^2 D^2\right] = (ML^{-3})(L^2T^{-2})(L^2) = MLT^{-2}$$

### ✓ Step 4: Consistency verdict

Both additive terms on the RHS have identical dimensions to the LHS force ( $MLT^{-2}$ ).

**Conclusion:** The formula is dimensionally homogeneous. ✓

### Aerodynamic pitching moment

The pitching aerodynamic moment  $M_o$  on an aircraft wing is a function of airspeed  $V$ , wing area  $A$ , chord  $C$ , and air density  $\rho$  such that:

$$M_o = \beta V^2 AC \rho$$

What are the dimensions of the coefficient  $\beta$ ?

**(Ans: Dimensionless,  $[\beta] = 1$ )**

## Problem 4.2: solution

### ➡ Step 1 & 2: Variable dimensions

**Target:** determine the dimensions of  $\beta$  in  $[M, L, T]$ .  
Primary dimensions of variables:

- Pitching moment:  
 $[M_o] = \text{Force} \times \text{Length} = ML^2T^{-2}$
- Airspeed:  $[V] = LT^{-1}$
- Wing planform area:  $[A] = L^2$
- Wing chord:  $[C] = L$
- Air density:  $[\rho] = ML^{-3}$

### ➡ Step 3: Dimensional algebra

Evaluate the RHS dimensional group multiplying  $\beta$ :

$$\begin{aligned}[V^2 AC \rho] &= (L^2 T^{-2})(L^2)(L)(ML^{-3}) \\ &= ML^{(2+2+1-3)}T^{-2} = \mathbf{ML^2T^{-2}}\end{aligned}$$

Isolating  $[\beta]$  from dimensional homogeneity:

$$[\beta] = \frac{[M_o]}{[V^2 AC \rho]} = \frac{ML^2T^{-2}}{ML^2T^{-2}} = 1$$

### ✓ Step 4: Sanity check

$[\beta] = 1$  (dimensionless).

### Falling sphere viscometer formulas

The following formulas were suggested to infer the fluid viscosity  $\mu$  and minimum fall time  $t$  for a sphere of net submerged weight  $W_{\text{net}}$ , diameter  $D$ , falling through distance  $L$ :

$$\mu \approx \frac{W_{\text{net}} t}{3\pi D L}, \quad t \geq \frac{2\rho D L}{\mu}$$

Are the above formulas dimensionally homogeneous?

**(Ans: Yes, both are dimensionally homogeneous)**

### ➡ Step 1 & 2: Dimensions of variables

Primary dimensions in  $[M, L, T]$ :

- Net weight (force):  $[W_{\text{net}}] = MLT^{-2}$
- Diameter & distance:  $[D] = [L] = L$
- Fall time:  $[t] = T$
- Fluid density:  $[\rho] = ML^{-3}$
- Dynamic viscosity:  $[\mu] = ML^{-1}T^{-1}$
- Pure numbers:  $[3\pi] = [2] = 1$

## Problem 4.3: solution

### ➡ Step 3: Verification of formulas

**Formula 1 (viscosity relation):**

$$\left[ \frac{W_{\text{net}} t}{3\pi DL} \right] = \frac{(MLT^{-2})(T)}{(L)(L)} = ML^{-1}T^{-1} = [\mu]$$

**Formula 2 (time limit relation):**

$$\left[ \frac{2\rho DL}{\mu} \right] = \frac{(ML^{-3})(L^2)}{ML^{-1}T^{-1}} = T = [t]$$

### ✓ Step 4: Conclusion

Both RHS expressions yield the exact primary dimensions of their respective target LHS variables.

**Both viscometer relations are dimensionally homogeneous. ✓**

# Problem 5

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## Problem 5: statement

A smooth-sided piston 5 cm long and of diameter 5.000 cm moves within a cylinder of 5.010 cm diameter. The lubricating oil has a viscosity of  $2 \times 10^{-3} \text{ N s m}^{-2}$ . What force is required to move the piston at a speed of  $12 \text{ m s}^{-1}$ ? Assume the velocity in the gap between the cylinder and the piston varies linearly, as indicated in the sketch.

**(Ans: 3.77 N)**

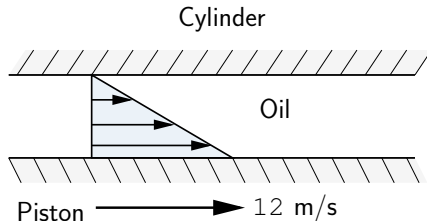


Figure: Linear velocity profile in lubrication clearance gap.

## Problem 5: solution (part 1)

### ➔ Step 1: Sketch & given data

#### Given parameters (piston in cylinder):

- Piston diameter:  $D = 5.000 \text{ cm} = 0.050 \text{ m}$
- Cylinder bore:  $D_c = 5.010 \text{ cm} = 0.0501 \text{ m}$
- Piston length:  $L = 5 \text{ cm} = 0.050 \text{ m}$
- Sliding velocity:  $U = 12 \text{ m s}^{-1}$
- Oil viscosity:  $\mu = 2 \times 10^{-3} \text{ N s m}^{-2}$

**Target:** driving force  $F$  to maintain steady speed.

### ➔ Step 2: Governing laws & assumptions

#### Newton's law of viscosity:

$$\tau = \mu \frac{du}{dy} = \mu \frac{U}{h}, \quad F = \tau A$$

#### Assumptions:

- **Narrow gap Couette shear:**  $h \ll D$ , so curvature effects are negligible and velocity varies linearly across  $h$ .
- **No-slip condition** at piston surface ( $u = U$ ) and cylinder bore ( $u = 0$ ).
- Constant Newtonian oil viscosity ( $\mu = \text{const}$ ).

## Problem 5: solution (part 2)

### ➡ Step 3: Algebra & symbolic derivation

Radial clearance gap  $h$ :

$$h = \frac{D_c - D}{2} = \frac{5.010 - 5.000}{2} \text{ cm} = 5 \times 10^{-5} \text{ m}$$

Wetted cylindrical shear area  $A$ :

$$A = \pi DL = \pi(0.050)(0.050) = 7.854 \times 10^{-3} \text{ m}^2$$

Closed symbolic force formula:

$$F = \tau A = \left( \mu \frac{U}{h} \right) (\pi DL)$$

### ✓ Step 4: Calculate & sanity check

Numerical evaluation:

$$\begin{aligned} F &= \left( 2 \times 10^{-3} \frac{12}{5 \times 10^{-5}} \right) (7.854 \times 10^{-3}) \\ &= (480 \text{ Pa})(7.854 \times 10^{-3} \text{ m}^2) = \mathbf{3.77 \text{ N}} \end{aligned}$$

Physical sanity check:

- Shear rate  $\dot{\gamma} = \frac{U}{h} = 2.4 \times 10^5 \text{ s}^{-1}$  is large.
- Small oil viscosity yields a very moderate, realistic friction force of 3.77 N.
- Units:  $[\text{Pa}][\text{m}^2] = \text{N}$ . ✓

# Problem 6

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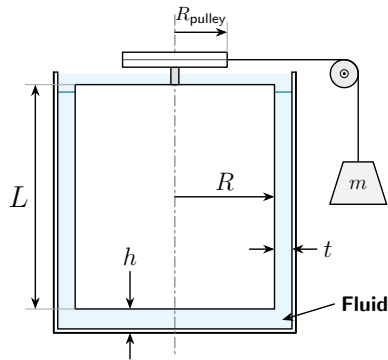
## Problem 6.1: statement

The drawing shows a concentric-cylinder viscometer. The fluid fills the gap between the two cylinders. The falling weight unreels a string from the pulley, thus rotating the inner cylinder. The viscosity of the fluid is then inferred from the rotational speed of the cylinder. Assume linear velocity profiles in both the radial gap and the flat bottom gap:

- Cylinder radius:  $R = 6 \text{ cm}$
- Pulley radius:  $R_{\text{pulley}} = 3 \text{ cm}$
- Cylinder length:  $L = 15 \text{ cm}$
- Clearances:  $t = h = 0.1 \text{ cm}$
- Falling weight mass:  $m = 30 \text{ g}$
- Steady rotation:  $N = 30 \text{ RPM}$

**Question 6.1:** Estimate the fluid viscosity  $\mu$  neglecting the frictional torque due to the fluid in the bottom gap.

**(Ans:  $1.381 \times 10^{-2} \text{ N s m}^{-2}$ )**



## Problem 6.1: solution (part 1)

### ➔ Step 1: Given parameters & kinematics

#### Viscometer dimensions & speed:

- Cylinder radius:  $R = 6 \text{ cm} = 0.06 \text{ m}$
- Cylinder length:  $L = 15 \text{ cm} = 0.15 \text{ m}$
- Radial gap:  $t = 0.1 \text{ cm} = 10^{-3} \text{ m}$
- Pulley radius:  $R_{\text{pulley}} = 3 \text{ cm} = 0.03 \text{ m}$
- Suspended mass:  $m = 30 \text{ g} = 0.030 \text{ kg}$
- Steady rotation:  $N = 30 \text{ RPM}$

#### Angular velocity:

$$\omega = \frac{30 \times 2\pi}{60} = \pi \approx 3.1416 \text{ rad s}^{-1}$$

### ➔ Step 2: Governing laws & torque balance

#### Steady rotational equilibrium:

$$\sum T_z = 0 \implies T_{\text{drive}} = T_{\text{viscous}}$$

#### Driving gravitational torque on pulley:

$$\begin{aligned} T_{\text{drive}} &= mgR_{\text{pulley}} \\ &= (0.030 \text{ kg})(9.81 \text{ m s}^{-2})(0.03 \text{ m}) \\ &= 8.829 \times 10^{-3} \text{ N m} \end{aligned}$$

#### Assumptions for 6.1:

- Narrow radial gap ( $t \ll R$ )  $\implies$  linear velocity profile.
- Neglect viscous resisting torque on the bottom disk gap.

## Problem 6.1: solution (part 2)

### ➔ Step 3: Cylindrical torque derivation

At surface  $r = R$ , tangential velocity is  $U = \omega R$ :

$$\tau = \mu \frac{du}{dr} \approx \mu \frac{\omega R}{t}$$

Total viscous torque on cylindrical wall ( $A = 2\pi RL$ ):

$$\begin{aligned} T_{\text{cyl}} &= \tau AR = \left( \mu \frac{\omega R}{t} \right) (2\pi RL) R \\ &= \frac{2\pi\mu\omega R^3 L}{t} \end{aligned}$$

Equating  $T_{\text{cyl}} = T_{\text{drive}}$  and solving for  $\mu$ :

$$\mu = \frac{\mathbf{T}_{\text{drive}} \mathbf{t}}{2\pi\omega \mathbf{R}^3 \mathbf{L}}$$

### ✓ Step 4: Calculate & sanity check

Substitute known values ( $R = 0.06 \text{ m}$ ,  $L = 0.15 \text{ m}$ ,  $t = 10^{-3} \text{ m}$ ):

$$\begin{aligned} \mu &= \frac{(8.829 \times 10^{-3} \text{ N m})(10^{-3} \text{ m})}{2\pi(\pi \text{ rad s}^{-1})(0.06 \text{ m})^3(0.15 \text{ m})} \\ &= \mathbf{1.381 \times 10^{-2} \text{ N s m}^{-2}} \end{aligned}$$

*Physical sanity check:*

- Value is  $\approx 13.8 \text{ mPa} \cdot \text{s} \approx 14 \text{ cP}$ , typical of a light motor oil or spindle lubricant (water is  $\approx 1 \text{ cP}$ ).
- Units:  $\frac{[\text{N m}][\text{m}]}{[\text{m}^4]} = \text{N s m}^{-2}$ . ✓

## Problem 6.2: statement

### Accounting for bottom gap shear

Repeat (6.1) to estimate the fluid viscosity  $\mu$ , but now include the resisting torque due to the fluid in the bottom gap between the flat bottom of the inner cylinder and the outer container.

**(Ans:**  $1.255 \times 10^{-2} \text{ N s m}^{-2}$ )

#### Bottom gap kinematics ( $h = 0.1 \text{ cm}$ )

The rotating base has radius  $R$ . Unlike the cylinder wall where velocity is constant ( $\omega R$ ), the base velocity varies linearly with radius:  $u(r) = \omega r$  for  $r \in [0, R]$ .

## Problem 6.2: solution (part 1)

### ➔ Step 2: Base kinematics & shear stress

**Linear velocity variation along base:** Unlike the cylinder wall where velocity is constant ( $\omega R$ ), the flat base velocity increases with radius:

$$u(r) = \omega r, \quad r \in [0, R]$$

**Local Newtonian shear stress:**

$$\tau(r) = \mu \frac{du}{dz} = \mu \frac{\omega r}{h}$$

where  $h = 0.1 \text{ cm} = 10^{-3} \text{ m}$  is the bottom clearance gap.

### ➔ Step 3: Resisting torque integration

On an elemental ring of area  $dA = 2\pi r dr$ :

$$\begin{aligned} dT_{\text{bottom}} &= r [\tau(r) dA] = r \left( \mu \frac{\omega r}{h} \right) (2\pi r dr) \\ &= \frac{2\pi\mu\omega}{h} r^3 dr \end{aligned}$$

Integrating across the disk from  $r = 0$  to  $r = R$ :

$$T_{\text{bottom}} = \frac{2\pi\mu\omega}{h} \int_0^R r^3 dr = \frac{\pi\mu\omega R^4}{2h}$$

## Problem 6.2: solution (part 2)

### ➡ Step 3: Total torque balance

Since  $t = h$ :

$$\begin{aligned} T_{\text{total}} &= T_{\text{cyl}} + T_{\text{bottom}} \\ &= \frac{2\pi\mu\omega R^3 L}{t} + \frac{\pi\mu\omega R^4}{2t} \\ &= \frac{\pi\mu\omega R^3}{t} \left( 2L + \frac{R}{2} \right) \end{aligned}$$

Solving symbolically for dynamic viscosity  $\mu$ :

$$\mu = \frac{\mathbf{T}_{\text{drive}} \mathbf{t}}{\pi\omega \mathbf{R}^3 (2\mathbf{L} + \mathbf{R}/2)}$$

### ✔ Step 4: Calculate & sanity check

Evaluating with  $2L + R/2 = 2(0.15) + 0.03 = 0.33 \text{ m}$ :

$$\begin{aligned} \mu &= \frac{(8.829 \times 10^{-3})(10^{-3})}{\pi(\pi)(0.06)^3(0.33)} \\ &= \mathbf{1.255 \times 10^{-2} \text{ N s m}^{-2}} \end{aligned}$$

*Pedagogical sanity check:*

- Neglecting the bottom gap in 6.1 gave  $\mu = 1.381 \times 10^{-2}$ , an overestimate of +10.0%  $\left( \frac{1.381 - 1.255}{1.255} \right)$ .
- Bottom gap end effects in rotational viscometers are non-negligible! ✔

# Problem 7

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## Problem 7: statement

The water strider insect below is supported on the water surface by surface tension. Assuming a saline water with density  $\rho$  and surface tension  $\gamma$ , insect weight  $W$  that is equally distributed over four legs only, and a water–insect leg interface tangent of angle  $\theta$  measured from the vertical, derive an expression for the minimum length of the water–insect leg interface,  $L_{\min}$ , to support its weight on the water surface. (Ans:  $W/(8\gamma \cos \theta)$ )

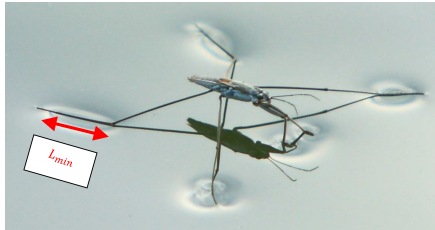


Figure: Water strider resting on free surface.

# Problem 7: solution (part 1)

## ➔ Step 1: Sketch & given data

### Given parameters (water strider):

- Total insect body weight:  $W$
- Weight equally distributed over 4 supporting legs:

$$W_{\text{leg}} = \frac{W}{4}$$

- Fluid properties: Saline water with density  $\rho$  and surface tension  $\gamma$
- Contact tangent angle:  $\theta$  measured from the vertical

**Target:** minimum interface length per leg  $L_{\text{min}}$ .

## ➔ Step 2: Governing laws & assumptions

### Static vertical equilibrium:

$$\sum F_y = 0 \implies W_{\text{leg}} = F_{\sigma,y}$$

**Key physical observation (contact lines):** Each leg depresses the fluid surface, forming **two distinct three-phase contact lines** along its wetted length (one along each side of the leg):

$$L_{\text{contact,total}} = 2 \times L_{\text{min}}$$

### Assumptions:

- Uniform contact angle  $\theta$  along the full length.
- Leg buoyancy is small compared to surface tension force.

## Problem 7: solution (part 2)

### ➔ Step 3: Algebra & symbolic derivation

**Vertical surface tension force per leg:** The vertical component of surface tension acting on both contact lines is:

$$F_{\sigma,y} = (2L_{\min}) \gamma \cos \theta$$

Equating upward surface tension to downward leg weight:

$$\frac{W}{4} = 2\gamma L_{\min} \cos \theta$$

Solving for the minimum interface length  $L_{\min}$ :

$$L_{\min} = \frac{W}{8\gamma \cos \theta}$$

### ✔ Step 4: Sanity check & physical limits

- **Superhydrophobic limit** ( $\theta \rightarrow 0^\circ$ ):  
 $\cos \theta \rightarrow 1 \Rightarrow L_{\min} = \frac{W}{8\gamma}$  (minimum required leg length for support).
- **Wetted limit** ( $\theta \rightarrow 90^\circ$ ):  
 $\cos \theta \rightarrow 0 \Rightarrow L_{\min} \rightarrow \infty$  (surface tension pulls purely horizontally; insect must sink).
- **Leg count generalization:** For  $N$  legs,  
 $L_{\min} = \frac{W}{2N\gamma \cos \theta}$ . For  $N = 4$ , the factor is  $2(4) = 8$ .
- **Dimensional check:**

$$[L_{\min}] = \frac{[W]}{[\gamma]} = \frac{MLT^{-2}}{MT^{-2}} = L \quad \checkmark$$

# Problem 8

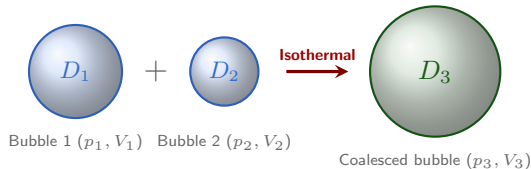
- 1 Quick reference
- 2 Problem 1
- 3 Problem 2
- 4 Problem 3

- 5 Problem 4
- 6 Problem 5
- 7 Problem 6
- 8 Problem 7
- 9 Problem 8

## Problem 8: statement

A soap bubble of diameter  $D_1$  coalesces with another bubble of diameter  $D_2$  to form a single bubble  $D_3$  with the same amount of air. For an isothermal process (constant temperature), express  $D_3$  as a function of  $D_1$ ,  $D_2$ ,  $p_{\text{atm}}$ , and surface tension  $\gamma$ .

**(Ans:**  $p_{\text{atm}}D_3^3 + 8\gamma D_3^2 = (p_{\text{atm}}D_1^3 + 8\gamma D_1^2) + (p_{\text{atm}}D_2^3 + 8\gamma D_2^2)$ )



## Problem 8: solution (part 1)

### ➡ Step 1 & 2: Laws & mass conservation

**Initial/final state:** Bubbles 1 ( $D_1$ ) and 2 ( $D_2$ ) coalesce into bubble 3 ( $D_3$ ) at  $T = \text{const.}$

**Young–Laplace (double interface thin film):**

$$\Delta P_i = \frac{4\gamma}{R_i} = \frac{8\gamma}{D_i}$$

$$P_i = p_{\text{atm}} + \frac{8\gamma}{D_i}$$

**Conservation of air mass (ideal gas,  $T = \text{const.}$ ):**

$$m_1 + m_2 = m_3 \implies P_1 V_1 + P_2 V_2 = P_3 V_3$$

### ➡ Step 3: Volume & pressure substitution

Spherical bubble volume in terms of diameter:

$$V_i = \frac{4}{3}\pi R_i^3 = \frac{\pi}{6} D_i^3$$

Substitute  $P_i$  and  $V_i$  into the mass conservation equation:

$$\sum_{i=1}^2 \left( p_{\text{atm}} + \frac{8\gamma}{D_i} \right) \frac{\pi}{6} D_i^3 = \left( p_{\text{atm}} + \frac{8\gamma}{D_3} \right) \frac{\pi}{6} D_3^3$$

Cancel the common factor  $\frac{\pi}{6}$ :

$$\sum_{i=1}^2 (p_{\text{atm}} D_i^3 + 8\gamma D_i^2) = p_{\text{atm}} D_3^3 + 8\gamma D_3^2$$

# Problem 8: solution (part 2)

## ✓ Step 4: Final coalescence relation

Expanding and rearranging terms:

$$p_{\text{atm}} D_3^3 + 8\gamma D_3^2 = (p_{\text{atm}} D_1^3 + 8\gamma D_1^2) + (p_{\text{atm}} D_2^3 + 8\gamma D_2^2)$$

**Physical interpretation:** The first term on each side represents the **internal energy / PV work** of the enclosed gas, while the second term represents the **surface energy** of the double liquid-gas interfaces!

## ➡ Asymptotic physical sanity checks

- **Large bubbles** ( $p_{\text{atm}} \gg \frac{8\gamma}{D}$  or  $\gamma \rightarrow 0$ ):

$$D_3^3 = D_1^3 + D_2^3$$

Standard volume conservation for incompressible fluids!

- **Zero external pressure** ( $p_{\text{atm}} \rightarrow 0$ ):

$$D_3^2 = D_1^2 + D_2^2$$

Surface area conservation!

- Physical limits behave with perfect consistency. ✓



**Questions?**